ONE OUT TRAINING FOR PHYSICS EMULATORS USING CONTINUOUS CONVOLUTIONS

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ABSTRACT

This report studies density prediction for Lagrangian, particle–based fluid emulators using a continuous–convolution (CConv) model in the SFBC style. We evaluate three datasets—SFBC Dataset II, LagrangeBench TGV–2D (converted to the SFBC format), and SFBC–TGV—and use density as a simple, per–particle target that reflects how well local interactions assemble into coherent flow structures. Our methodology compares spectral kernel parameterizations (Fourier variants, Chebyshev), varies message–passing depth to probe receptive–field effects, and quantifies compute–accuracy trade-offs via FLOPs. Motivated by spatial artifacts observed on LagrangeBench, we introduce a correlation-based evaluation and a hybrid loss (MSE plus a Pearson-correlation term).

On LagrangeBench TGV–2D, correlation aligns closely with visual quality, and the hybrid loss consistently increases correlation while often reducing MSE—most notably for shallow–to–mid-depth Fourier models—leading to sharper and more stable density fields. Baselines on SFBC Dataset II and SFBC–TGV behave as expected and serve to anchor comparisons across resolutions and generation schemes. We also test adding fluid features and find modest gains for weaker bases while strong even-Fourier configurations remain largely unchanged, consistent with CConv's use of relative geometry.

Overall, the study indicates that pairing spectral inductive biases with moderate depth and a correlation-aware objective improves spatial fidelity without additional inference cost. We conclude with recommendations for broader evaluations (e.g., 3D flows, richer architectures) and for testing the hybrid objective on more complex regimes such as high-resolution SFBC–TGV.

1 INTRODUCTION

Learning in Lagrangian, particle-based fluid simulations is demanding. The state lives on moving particles whose neighborhoods change over time, so the interaction graph is dynamic and local errors can compound over long rollouts. Effective models also need inductive biases that respect symmetries and conservation laws while remaining computationally efficient at large particle counts (Wang et al. (2024)). In this study, we focus on density as the central target. It is a per-particle quantity used throughout SPH workflows, drives pressure forces and compressibility control, and offers a direct readout of whether learned local interactions assemble into coherent flow structures (Koschier et al. (2022)). Treating density prediction this way lets us compare surrogates across datasets without entangling the task with downstream integration choices.

Our study connects two complementary perspectives on this problem. SFBC (Winchenbach & Thuerey (2024)) frames learning as estimating particle-interaction operators via symmetric basis convolutions and reports strong results on SPH test cases, while LagrangeBench (Toshev et al. (2023)) defines a temporally coarse-grained Lagrangian prediction task with long rollouts and evaluates surrogates using physical metrics beyond pointwise errors. Motivated by the spatial nature of errors observed on LagrangeBench, we apply the SFBC network to three datasets (SFBC Dataset II, LagrangeBench TGV-2D, and SFBC-TGV) and introduce a correlation-based evaluation and a hybrid loss aimed at improving the spatial alignment of predicted density fields in that setting.

In brief, we use Pearson correlation (Lee (2001)) to assess spatial agreement and incorporate it into training as a hybrid objective where LagrangeBench TGV-2D experiments indicate improved spatial

alignment. We complement this with analyses of receptive-field size and feeding additional fluid-feature to the network. The remainder of the report reviews background and methodology, details datasets, presents experiments with and without the correlation term, and closes with limitations and next steps.

2 Background and Related Work

We will describe smoothed particle hydrodynamics, summarize the SFBC approach and its training objective, outline LagrangeBench with the 2D Taylor–Green vortex (TGV) (Taylor & Green (1937)) dataset we use, recall the TGV flow itself, and review Pearson correlation as a spatial similarity measure.

2.1 Smoothed Particle Hydrodynamics (SPH)

SPH represents a continuum by particles of mass m_i at positions $\mathbf{x}_i$ (Monaghan (2012)). Field quantities are estimated by kernel-weighted neighbor sums over a compact support of radius h. For density,

$$\rho_i = \sum_j m_j \, W\big(\|\mathbf{x}_i - \mathbf{x}_j\|, h\big), \tag{1}$$

where $W(\cdot, h)$ is a smoothing kernel with finite support "support radius" and typical cutoff $r \leq 2h$.

2.2 Symmetric Fourier Basis Convolutions for Particle-Based Fluid Learning (SFBC)

SFBC formulates particle interactions via continuous, symmetry-aware basis convolutions (e.g., even/odd Fourier bases) to learn surrogate operators for Lagrangian flows, reporting robust accuracy on SPH-style tasks (Winchenbach & Thuerey (2024)). During supervised training, the main objective is a per-sample mean-squared error (MSE)

$$\mathcal{L}_{\text{SFBC}} = \frac{1}{N} \sum_{i=1}^{N} \|\hat{\mathbf{z}}_i - \mathbf{z}_i\|_2^2 \,, \tag{2}$$

where $\mathbf{z}_i$ denotes the ground-truth target (e.g., a per-particle or grid-mapped field such as density) and $\hat{\mathbf{z}}_i$ the prediction; this loss penalizes pointwise discrepancies and is complemented at evaluation by grid-resampled field errors and spectral comparisons.

2.3 LagrangeBench: A Lagrangian Fluid Mechanics Benchmarking Suite

LagrangeBench provides curated SPH-generated trajectories with *coarse-grained* prediction protocol in which models learn to predict particle displacements at a fixed stride (e.g., every 100 simulator steps) (Toshev et al. (2023)). Evaluation includes *field-based errors*: particle quantities are resampled to a regular grid and compared as fields (e.g., PSNR, L_2, spectral energy), complementing per-particle metrics. In this study, the selected dataset from the suite is the 2D Taylor–Green vortex (TGV), a canonical vortical flow used to probe coherence and decay behavior. Conceptually, this task differs from SFBC in that LagrangeBench trains models to map states to future positions, while SFBC centers on learning kernel-like interaction operators whose outputs are compared via field-based errors.

2.4 The Two-Dimensional Taylor–Green Vortex (TGV)

The 2D TGV models the decay of a large-scale, periodic vortex lattice into progressively weaker structures under viscosity on a periodic domain, serving as a paradigmatic test of vortex dynamics and energy decay (Taylor & Green (1937)). On a square domain of side $2\pi/k$, the equation evolves (in the incompressible viscous case) with an exponentially decaying amplitude,

$$u(x, y, t) = U_0 \, e^{-2\nu k^2 t} \, \sin(kx) \cos(ky), \tag{3}$$

$$v(x, y, t) = -\, U_0 \, e^{-2\nu k^2 t} \, \cos(kx) \sin(ky), \tag{4}$$

leading to exponential kinetic-energy decay and gradual smoothing of vortical structures.

2.5 PEARSON CORRELATION FOR SPATIAL EVALUATION

Pearson's correlation coefficient quantifies linear association between two scalar fields while being invariant to affine rescaling, making it useful for judging spatial similarity of predicted and reference maps (e.g., density fields) beyond pure amplitude matching:

$$\operatorname{corr}(\hat{\boldsymbol{\phi}}, \boldsymbol{\phi}) = \frac{\sum_{i=1}^{M} \left(\hat{\phi}_i - \overline{\hat{\phi}}\right)\left(\phi_i - \overline{\phi}\right)}{\sqrt{\sum_{i=1}^{M} \left(\hat{\phi}_i - \overline{\hat{\phi}}\right)^2} \sqrt{\sum_{i=1}^{M} \left(\phi_i - \overline{\phi}\right)^2}}. \tag{5}$$

It is widely used in spatial statistics (Lee (2001)), image analysis (Hoque et al. (2024)), and geoscience (Lin et al. (2021)) to assess pattern agreement between fields where structural alignment (rather than absolute scale) is critical.

3 METHODOLOGY

The following subsections outline the modeling approach, inductive biases, and the hybrid loss formulation that guide our experiments.

3.1 GRAPH NEURAL NETWORKS

GNNs model fluids in a Lagrangian view where particles are graph nodes and edges connect neighbors within a cutoff radius; edge features are built from relative geometry (e.g., $\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i$ and its radial encoding), while node features carry fluid attributes (e.g., density, velocity) and optional constants for ablations. A message-passing block aggregates edge messages with local MLPs and updates node states (Sanchez-Gonzalez et al. (2020)), and stacking L such blocks yields a receptive field of roughly L times the neighbor radius; mirroring how SPH operators are local and composed over neighborhoods. This design is translation-invariant (and permutation-invariant by construction), aligns with SPH's reliance on relative positions, and serves as a common substrate to compare architectural choices and receptive-field depth across datasets. A single block therefore approximates an SPH-style kernel sum (Eq. 1); stacking L blocks composes these local interactions over L neighborhoods, effectively enlarging the receptive field.

3.2 SPECTRAL BASIS PARAMETERIZATIONS (FOURIER/CHEBYSHEV) AS INDUCTIVE BIASES

To encode symmetry and smoothness in particle interactions, we parameterize continuous convolution kernels with low-dimensional spectral bases. Following the SFBC idea of symmetric basis filters, we use even/odd Fourier components to capture parity and isotropy (Li et al. (2020)) and Chebyshev polynomials to provide stable radial function families on compact intervals (Winchenbach & Thuerey (2024)). Kernels are learned as linear combinations of these bases and evaluated on relative offsets between particles. In practice, this yields compact models with controllable receptive fields, good sample efficiency, stable gradient behavior, and it integrates cleanly with message-passing by treating basis-evaluated edge features as inputs to edge MLPs or continuous convolution layers.

3.3 LOSSES: SFBC BASELINE, CORRELATION METRIC, AND HYBRID OBJECTIVE

The SFBC training objective we adopt is a pointwise mean-squared error (MSE) between predictions and targets (e.g., density), which we treat as the baseline criterion. Motivated by spatial artifacts observed on the LagrangeBench TGV-2D task after converting it to the SFBC format, we add a correlation-based loss term to measure spatial agreement between predicted and reference density fields as

$$\mathcal{L}_{\text{corr}} = 1 - \frac{1}{B} \sum_{b=1}^{B} \operatorname{corr}\big(\hat{\rho}^{(b)}, \rho^{(b)}\big), \tag{6}$$

Because correlation is scale-invariant, a model could rescale amplitudes yet achieve a high score. To discourage such trivial rescalings we include a light penalty on mean shifts:

$$\mathcal{L}_{\text{scale}} = \frac{1}{B} \sum_{b=1}^{B} \left(\frac{\bar{\hat{\rho}}^{(b)} - \bar{\rho}^{(b)}}{\bar{\rho}^{(b)} + \varepsilon} \right)^2, \tag{7}$$

with ε preventing division by zero. Finally, we combine terms into a hybrid objective

$$\mathcal{L}_{\text{hyb}} = \alpha \, \mathcal{L}_{\text{SFBC}} + \beta \, \mathcal{L}_{\text{corr}} + \lambda \, \mathcal{L}_{\text{scale}}, \tag{8}$$

where $\mathcal{L}_{\text{SFBC}}$ denotes the baseline MSE used by SFBC. In our study, this hybrid loss improved spatial alignment on LagrangeBench TGV-2D, particularly benefiting shallower models; we will report comparisons across receptive fields and fluid-feature choices (e.g., relative distance) in the experiments section and extend the analysis to the other datasets.

4 DATASETS

We evaluate on three Lagrangian SPH datasets prepared in a unified SFBC trainable format: SFBC Dataset II, the 2D Taylor–Green Vortex (TGV) from LagrangeBench, and a SFBC–TGV set obtained by chunking high-resolution trajectories. This section describes the formatting choices, summarize density statistics, and report Earth Mover Distance (EMD) as a compact measure of temporal change in particle configurations.

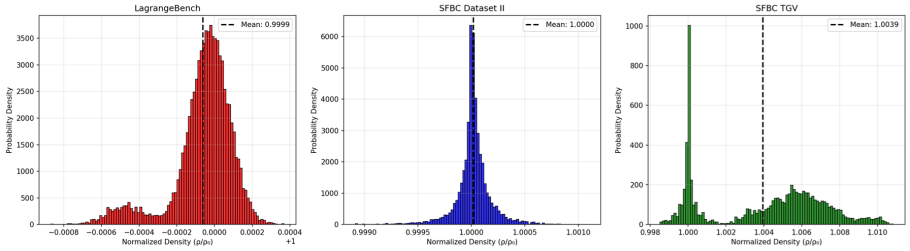


Figure 1: Density distributions ρ normalized on ρ_0 (density at rest) aggregated over all frames for the three datasets. The dashed vertical line marks the mean of each distribution, where all datasets are centered near 1.

4.1 SFBC DATASET II

Formatting: We use the dataset as provided in the SFBC study and consume it directly with the SFBC pipeline; the simulation employs weakly-compressible SPH on a square domain with mixed boundaries (Winchenbach & Thuerey (2024)). Each frame contains $N = 4096$ fluid particles and 804 boundary particles on the domain $[-1, 1]^2$; particles are placed on a regular lattice with spacing $dx \approx 0.03125$. The SPH support radius is $h = 0.0795$, and time integration uses a fixed step of $\Delta t = 10^{-3}$ s. Trajectories span 3200 frames; we use 36 for training and 16 for testing. Across all frames, normalized density is tightly concentrated around unity with mean ≈ 1.0000 and standard deviation $\approx 5.8 \times 10^{-5}$.

Density Distribution: The normalized density (ρ/ρ_0) histogram is sharply peaked and well-approximated by a narrow Gaussian centered at 1, with negligible skew or secondary modes. This reflects the weak compressibility and steady statistical behavior of the set.

Earth Mover Distance: For each trajectory; we take the middle frame, compute EMD from the start to middle and middle to last frames, and report the average over trajectories/splits (EMD compares particle configurations as distributions rather than index-matched points). In this dataset, the averaged EMD is moderate (on the order of a few $\times 10^{-3}$), i.e., larger than LagrangeBench TGV and smaller than SFBC–TGV. This reflects meaningful but smooth configuration changes over long sequences.

4.2 LAGRANGEBENCH TGV–2D

Formatting: The original benchmark logs positions every 100 solver steps (temporal coarse-graining) (Toshev et al. (2023)). We convert each trajectory to the SFBC frame-centric format and compute the missing SPH fields needed by SFBC (per-frame density via SPH summation, particle areas from dx, and support radii from metadata) while preserving the unit-square periodic domain and the original train/test splits. Each sequence contains $N = 2500$ particles evolving on $[0, 1]^2$; the particle spacing is $dx \approx 0.02$, and neighborhoods are built with a radius consistent with an SPH support of $= 0.029$. The simulator advances with internal steps of $\delta t = 4 \times 10^{-4}$ s and writes a frame every 100 steps, yielding an effective snapshot interval of $\Delta t = 0.04$ s. Trajectories have 126 frames; we use 100 for training and 50 for testing (we omit the original validation split to better match the SFBC Dataset II size). Normalized density remains close to one with a slightly larger spread than SFBC Dataset II (mean ≈ 0.99997, std $\approx 1.42 \times 10^{-4}$).

Density Distribution: The normalized density (ρ/ρ_0) distribution is not exactly Gaussian; it is narrowly centered near 1 but exhibits faint secondary shoulders and mild asymmetry, consistent with the coarse-grained sampling (every 100 SPH steps) and the resulting mixture of local states.

Earth Mover Distance: Using the same averaging protocol; LagrangeBench TGV exhibits the smallest temporal EMD among the three datasets (roughly $\sim 10^{-3}$), reflecting modest distributional drift between the selected frames under the coarse-grained sampling.

4.3 SFBC–TGV

Formatting: The original data contains four long ~ 4096 frame trajectories. Unlike SFBC Dataset II and LagrangeBench TGV, these trajectories were simulated with δ^+-SPH using an Euler time integration together with density-based particle shifting corrections, which promotes uniform particle distributions and near-incompressibility in the released fields. We chunk 4 long simulated trajectories into contiguous 128 frame sequences, then split into 85 train and 43 test trajectories (no data loss), where we randomly shuffle sub-trajectories into test and train sets. Simulations are 2D SPH on a near-unit square domain $[-0.496, 0.496]^2$ with $N = 16{,}384$ particles; the particle spacing is $dx \approx 7.75 \times 10^{-3}$ and the support radius $h = 0.02965$. Time integration uses a fixed step of $\Delta t = 10^{-3}$ s. In the released fields the flow is effectively incompressible (mean $= 1.0000$, std ≈ 0), while the higher resolution introduces richer small-scale structure than in the other sets.

Density Distribution: The normalized density (ρ/ρ_0) histogram is distinctly non-Gaussian and multi-modal, with several narrow peaks around values close to (but slightly above) unity. This indicates coexisting flow structures and a broader mixture of local configurations at the higher resolution.

Earth Mover Distance: Averaged EMD is the largest of the three (order 10^{-2}), consistent with faster spatial reorganization between the selected time windows at this resolution and flow regime.

5 EXPERIMENTS AND RESULTS

This section evaluates density prediction across SFBC Dataset II, LagrangeBench TGV–2D, and SFBC–TGV using the SFBC network. We first establish baselines under the original SFBC MSE objective, then study how model depth and message-passing receptive field influence accuracy (particularly on LagrangeBench) before introducing a Pearson based correlation metric and a hybrid loss to encourage spatial agreement, quantify compute–accuracy trade-offs via FLOPs versus loss/correlation, and assess the effect of fluid features.

5.1 DENSITY PREDICTION EXPERIMENTS

All density models are trained with the SFBC mean–squared error (MSE) objective on normalized densities. We follow the SFBC "constant input" setting (each particle carries a scalar 1 so the network can learn an $m\,W_{ij}$–like aggregation), and vary (i) message–passing depth, (ii) basis family (linear, Chebyshev, Fourier, even/odd Fourier), and (iii) hidden widths. Curves are reported as

Dataset	Architecture	Basis	MSE	PSNR (dB)
SFBC Dataset II	[1]	fourier	6.46×10^{-5}	42.22
SFBC Dataset II	[1]	fourier_even	6.63×10^{-5}	41.99
SFBC Dataset II	[64,32,16,1]	fourier	4.60×10^{-5}	44.01
SFBC Dataset II	[64,32,16,1]	fourier_even	4.43×10^{-5}	43.73
LagrangeBench TGV–2D	[1]	fourier	5.65×10^{-4}	37.31
LagrangeBench TGV–2D	[32,16,8,1]	fourier_even	2.67×10^{-5}	50.62
LagrangeBench TGV–2D	[64,32,16,1]	fourier	3.19×10^{-5}	49.82
LagrangeBench TGV–2D	[64,32,16,1]	fourier_even	5.16×10^{-5}	53.80
SFBC–TGV	[1]	fourier_even	1.70×10^{-5}	47.23
SFBC–TGV	[1]	linear	3.70×10^{-5}	44.48
SFBC–TGV	[1]	chebyshev	1.46×10^{-4}	38.49

Table 1: Density prediction with the SFBC MSE objective on datasets: SFBC Dataset II, LagrangeBench TGV–2D, and SFBC–TGV.

training MSE (log–scale) and PSNR over 1000 iterations for representative runs; the qualitative statements below are consistent across repeats.

SFBC Dataset II This dataset serves to verify the pipeline and calibrate architectures by comparing with the example training behavior from SFBC. Increasing capacity from `[1]` to `[64,32,16,1]` lowers MSE by **28.8%** for the Fourier basis and by **33.2%** for even–Fourier (Table 1), with PSNR gains of **+1.79 dB** and **+1.74 dB**, respectively. Training curves consistently show faster convergence and better final accuracy for Fourier–family models than for linear/Chebyshev.

LagrangeBench TGV–2D We sweep depths and widths (11 variants in total, from `[1]` up to `[64,32,16,1]` with 1–5 message-passing layers). Two robust patterns emerge. *(i) Depth matters.* One message-passing layer plateaus at a substantially higher MSE and lower PSNR. Adding depth consistently reduces loss and improves PSNR, with marked gains by 3–4 layers. The shallow `[1]` baseline ends at an MSE of 5.65×10^{-4} with PSNR 37.31 dB, while moving to `[32,16,8,1]` (even–Fourier) reduces MSE by **95.3%** (to 2.67×10^{-5}) and raises PSNR by **+13.31 dB**; `[64,32,16,1]` (Fourier) still yields a **94.4%** MSE reduction and **+12.51 dB** (Table 1). The lowest MSE in our sweep appears at 3 layers for even–Fourier. *(ii) Spectral bases help:* at comparable capacity, even–Fourier/Fourier converge faster and to significantly lower loss than linear/Chebyshev. Because the SFBC loss is unitless on normalized density, absolute values are hard to interpret across datasets. Visual inspections show that models with lower MSE are not always the ones with the best spatial alignment of structures (causing unsatisfactory visual predictions), which motivates introducing a correlation based metric and hybrid loss later..

SFBC–TGV This set probes the same flow class as LagrangeBench at higher resolution. Under the `[1]` architecture, even–Fourier outperforms linear by **54%** lower MSE and **+2.75 dB** PSNR, and outperforms Chebyshev by **88%** lower MSE and **+8.74 dB** PSNR (Table 1), indicating that the advantage of spectral bases is not peculiar to SFBC Dataset II. and consistent with the TGV problem. We retain this baseline to anchor later comparisons and to isolate effects specific to the LagrangeBench coarse-grained setting.

5.2 Receptive Field Analysis

For a message–passing network with neighbor (support) radius r_{supp} and N CConv blocks, the effective receptive field grows approximately linearly, $r_{\text{eff}} \approx N\, r_{\text{supp}}$. In our LagrangeBench setup $r_{\text{supp}} \approx 0.029$ on $[0,1]^2$, giving $r_{\text{eff}} \approx$ (0.029, 0.058, 0.087, 0.116, 0.145) for $N = 1 \dots 5$. Because SPH density is computed by a compact kernel sum (Sec. 2.1), a single message-passing step *under-covers* the physical neighborhood required for accurate density; stacking 3–4 blocks expands r_{eff} past the kernel's influence region and supplies the missing context. This aligns with Table 1: deepening from `[1]` to `[32,16,8,1]` cuts MSE by 95% and boosts PSNR by 13 dB on LagrangeBench.

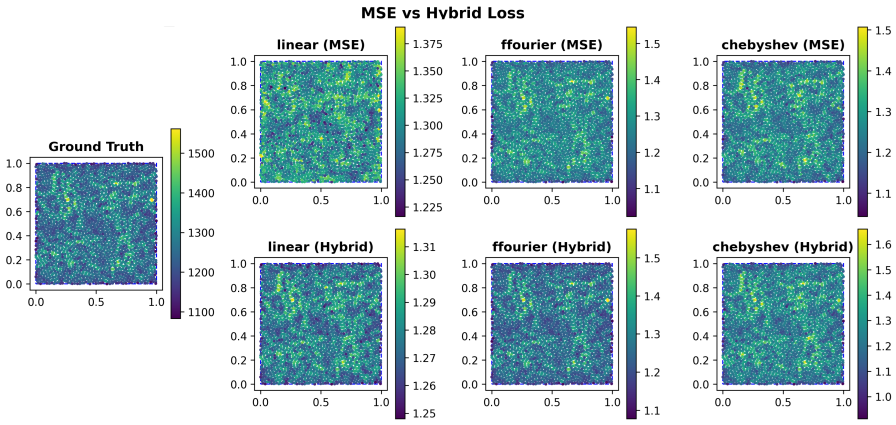


Figure 2: Qualitative comparison of MSE vs. Hybrid loss on LagrangeBench TGV–2D (test trajectory 15, frame 100). Left column shows the ground-truth density; the top row shows predictions from the `[32,16,1]` architecture trained with the plain MSE objective, and the bottom row shows the same architecture trained with the Hybrid objective (MSE + correlation + scale penalty).

Architecture	MSE — Corr	MSE — Corr (Hybrid)	FLOPs (M)
Linear			
linear [1]	2.17×10^{-2} — -0.002	3.87×10^{-2} — 0.206	6
linear [32, 16, 1]	6.89×10^{-3} — 0.498	7.12×10^{-3} — 0.916	601
linear [32, 16, 8, 1]	6.43×10^{-3} — 0.533	1.48×10^{-3} — 0.927	723
linear [64, 32, 16, 1]	5.57×10^{-3} — 0.704	8.24×10^{-4} — 0.962	2718
Chebyshev			
chebyshev [1]	2.58×10^{-2} — -0.198	4.95×10^{-2} — 0.037	10
chebyshev [32, 16, 1]	2.17×10^{-3} — 0.882	3.27×10^{-3} — 0.931	609
chebyshev [32, 16, 8, 1]	2.54×10^{-3} — 0.887	2.15×10^{-3} — 0.958	733
chebyshev [64, 32, 16, 1]	1.94×10^{-3} — 0.906	9.84×10^{-4} — 0.969	2728
Fourier			
fourier [1]	5.65×10^{-4} — 0.978	4.53×10^{-4} — 0.998	16
fourier [32, 16, 1]	8.66×10^{-4} — 0.955	4.82×10^{-5} — 0.999	621
fourier [32, 16, 8, 1]	4.39×10^{-4} — 0.976	2.65×10^{-5} — 0.999	748
fourier [64, 32, 16, 1]	3.19×10^{-5} — 0.998	4.01×10^{-5} — 0.999	2743

Table 2: For four architectures per basis, per-model forward cost (FLOPs; millions) and accuracy on LagrangeBench TGV–2D under the plain SFBC MSE objective and the hybrid objective.

Lower MSE does not always yield better density prediction visuals. With normalized densities, MSE mainly rewards amplitude accuracy and penalizes a few large outliers; it can be insensitive to phase shifts or localized blurring of coherent vortical patterns. As discussed in Sec. 2.3, frames are temporally coarse-grained; so salient structures can advect by multiple particle spacings between snapshots. With shallow message-passing, even when it reduces MSE, it can miss the spatial alignment of those patterns. This motivates augmenting MSE with a spatial agreement term (correlation; Sec. 3.3), which we show improves alignment and visual fidelity in the next section.

5.3 FLOPs and Hybrid Loss Analysis

Throughout, *FLOPs* denotes floating–point *operations per forward pass*. Our count includes a single neighbor search at the first layer, per–edge distance computations, basis evaluations, continuous-convolution multiplies/adds (scaling with edges and basis tensor products), scatter (aggregation), and activations. This hardware-agnostic measure lets us compare accuracy *vs.* compute.

Across architectures, the hybrid objective consistently increases correlation and often reduces MSE, with the effect strongest for Fourier models at shallow–mid depth and for deeper linear configurations (Table 2). For shallow non-spectral settings (e.g., `linear [1]`, `chebyshev [1]`), the

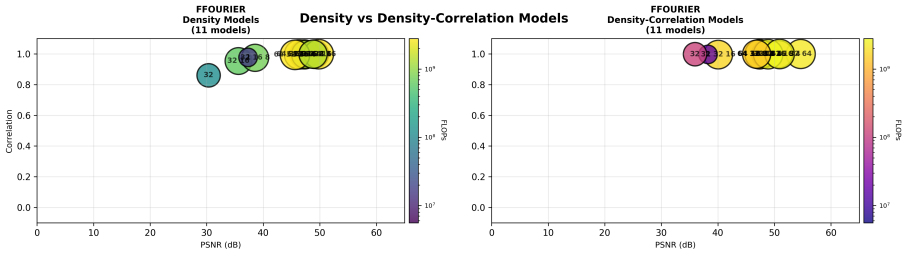


Figure 3: LagrangeBench TGV–2D, Fourier basis: PSNR vs. correlation for 11 architectures. Left: models trained with the MSE objective; right: models trained with the hybrid (MSE + correlation) objective. Each bubble is one model; the x–axis shows PSNR (dB) and the y–axis shows Pearson correlation. Bubble *brihtness* and bubble *area* is proportional to FLOPs.

hybrid pushes correlation up sharply (from near zero or negative to 0.21 and 0.04), while MSE increases. This trade-off shows the hybrid term is effectively prioritizing spatial alignment when amplitude calibration is weak, exactly the failure mode that motivated our objective. Among Fourier models, `[32,16,8,1]` reaches saturated correlation at $\sim$748 M FLOPs (about **3.7**$\times$ fewer operations than the largest `[64,32,16,1]` ($\sim$2743 M)) indicating a favorable operating point on the Pareto frontier. Qualitatively, *Figure 2* shows why this matters: hybrid training yields visibly crisper structures and better alignment of vortical patterns, which correlates with the large correlation gains reported in Table 2. Complementarily, the bubble plot in *Figure 3* shows most models shifting *up and to the right* under the hybrid loss (higher correlation and PSNR) at essentially the same compute, demonstrating that alignment can be improved without paying extra FLOPs.

5.4 Feeding Additional Fluid Features

We augument inputs with relative distance (in addition to the constant "1") and train informed models at the same depth as our 4-layer baselines using `[32,16,1]`. On MSE-only training, adding relative distance offers *modest* gains where the basis is weaker (linear and odd-Fourier) and leaves the strongest configuration (even–Fourier) essentially unchanged—consistent with CConv already operating on relative offsets. With the hybrid objective, informed models follow the same trend as their non-informed counterparts but at slightly higher performance: correlation quickly approaches 0.99 for Fourier variants and PSNR improves by roughly 1–3 dB over the informed MSE-only runs. With the improved correlation, we can still observe significant quality improvements on the visual predictions even if the PSNR gain is minimal. These results suggest complementarity rather than redundancy: extra relative-position cues assist narrow bases, while the hybrid loss reliably boosts spatial alignment without sacrificing pointwise error.

6 Conclusion and Future Work

We evaluated density prediction across SFBC Dataset II, LagrangeBench TGV-2D, and SFBC-TGV with an SFBC-style CConv model. On LagrangeBench TGV-2D, Pearson correlation tracks perceived visual quality better than MSE, and adding it to the loss (hybrid objective) improves spatial fidelity and training stability; correlation rises across architectures and depths, with often decrase in MSE. Together with our receptive-field analysis, these results support pairing spectral inductive biases with moderate depth and correlation-aware training for Lagrangian density inference at no extra inference cost.

Looking ahead; (i) **3D flows:** extend to three-dimensional benchmarks to test robustness under larger neighborhoods and more complex vortex dynamics, (ii) **Architectures:** explore deeper CConv stacks, equivariant/message-passing hybrids, learned or adaptive supports, and lightweight edge attention, and revisit feature design with normalized relative cues at shallow depth, (iii) **Datasets and tasks:** evaluate the hybrid objective on high-resolution SFBC-TGV (including $\dot{\rho}$ supervision and multi-step rollouts) and broaden beyond density to position/velocity with physically informed metrics, to identify where correlation-augmented training yields the largest gains and to carry the observed stability to more demanding regimes.

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